Lab 7 - Fuzz Testing Report

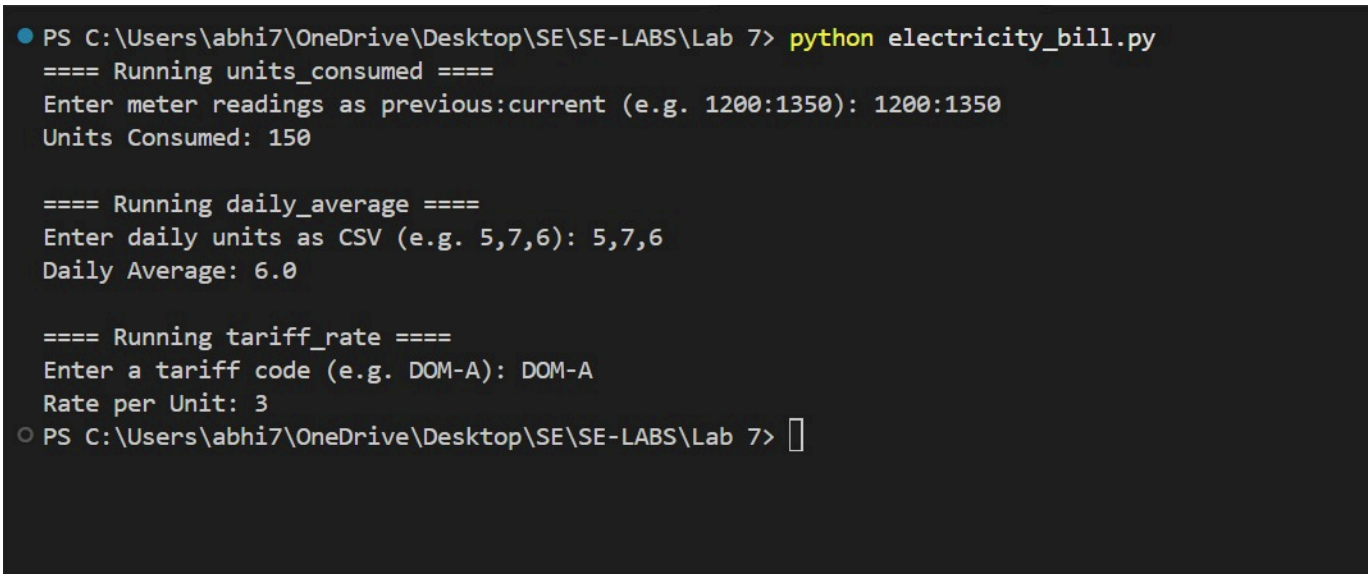
Electricity Bill Application | Hypothesis and pytest

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This report documents the original application execution, failures discovered through property-based fuzz testing, the input-validation fixes, and the final test results.

1. Original application execution

The original application was run with valid sample inputs: 1200:1350, 5,7,6, and DOM-A.



2. Initial Hypothesis failures

The initial fuzz tests exposed failures in all three functions. The detailed output shows minimized counterexamples and the exceptions raised.

```
PS C:\Users\abhi7\OneDrive\Desktop\SE\SE-LABS\Lab 7> python -m pytest test_electricity_bill.py -v
============================= test session starts =============================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\abhi7\AppData\Local\Python\pythoncore-3.14-64\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\abhi7\OneDrive\Desktop\SE\SE-LABS\Lab 7
plugins: hypothesis-6.170.14
collected 3 items

test_electricity_bill.py::test_units_consumed_no_crash FAILED                [ 33%]
test_electricity_bill.py::test_daily_average_no_crash FAILED                 [ 66%]
test_electricity_bill.py::test_tariff_rate_no_crash FAILED                   [100%]
```

```
_____ test_daily_average_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\abhi7\OneDrive\Desktop\SE\SE-LABS\Lab 7\test_electricity_bill.py", line 11, in test_daily_average_no_crash
  |     def test_daily_average_no_crash(value):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\abhi7\OneDrive\Desktop\SE\SE-LABS\Lab 7\test_electricity_bill.py", line 12, in test_daily_average_no_crash
    |     daily_average(value)
    |     ~~~~~~~~~~~~~~^^^^^^
    |   File "C:\Users\abhi7\OneDrive\Desktop\SE\SE-LABS\Lab 7\electricity_bill.py", line 9, in daily_average
    |     units = [int(u) for u in units_csv.split(',')]
    |              ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_daily_average_no_crash(
    |     value='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\abhi7\OneDrive\Desktop\SE\SE-LABS\Lab 7\test_electricity_bill.py", line 12, in test_daily_average_no_crash
    |     daily_average(value)
    |     ~~~~~~~~~~~~~~^^^^^^
    |   File "C:\Users\abhi7\OneDrive\Desktop\SE\SE-LABS\Lab 7\electricity_bill.py", line 9, in daily_average
    |     units = [int(u) for u in units_csv.split(',')]
    |                             ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_daily_average_no_crash(
    |     value=None,  # or any other generated value
    | )
    +------------------------------------
```

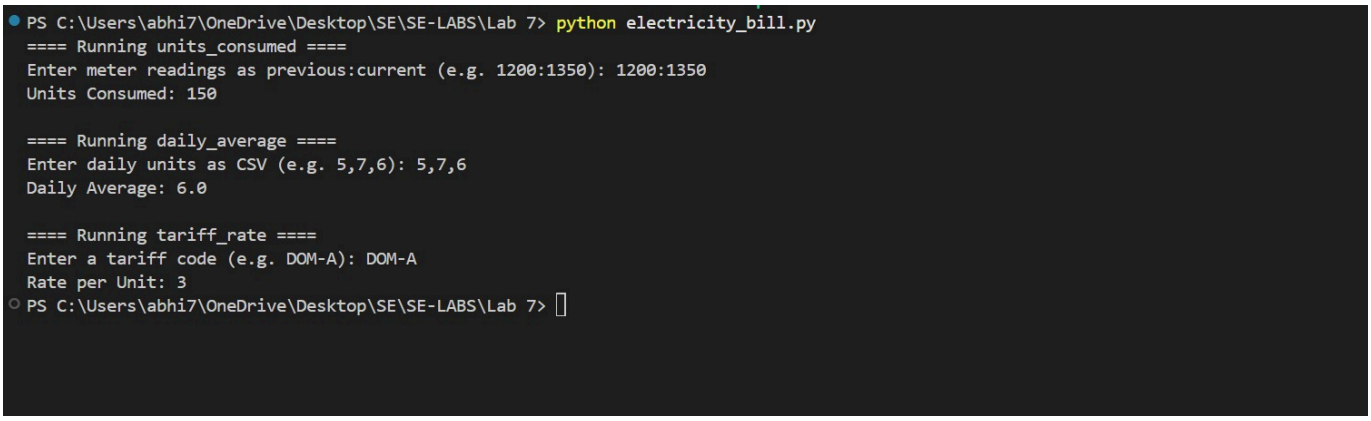
3. Initial failure summary

Pytest summary showing that all three initial fuzz-test functions failed.

```
=========================== short test summary info ===========================
FAILED test_electricity_bill.py::test_units_consumed_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_electricity_bill.py::test_daily_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_electricity_bill.py::test_tariff_rate_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 2.65s ==============================
PS C:\Users\abhi7\OneDrive\Desktop\SE\SE-LABS\Lab 7>
```

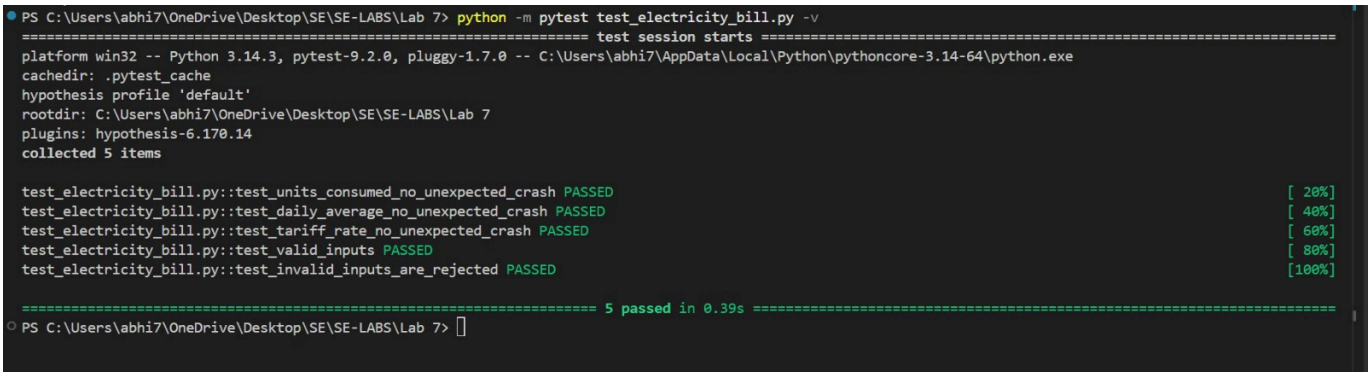
4. Fixed application execution

After input validation was added, the application returned the expected results for the same valid sample inputs.



5. Final Hypothesis test results

The updated test suite completed successfully: 5 tests passed.



Summary of findings

Hypothesis identified malformed meter readings, empty or invalid daily-unit CSV values, and invalid tariff codes or missing values. The application was updated to validate inputs and raise controlled ValueError exceptions for rejected inputs.

Final verification: 5 tests passed.